

SureClinical

SureClinical eTMF Cloud Compliance White Paper

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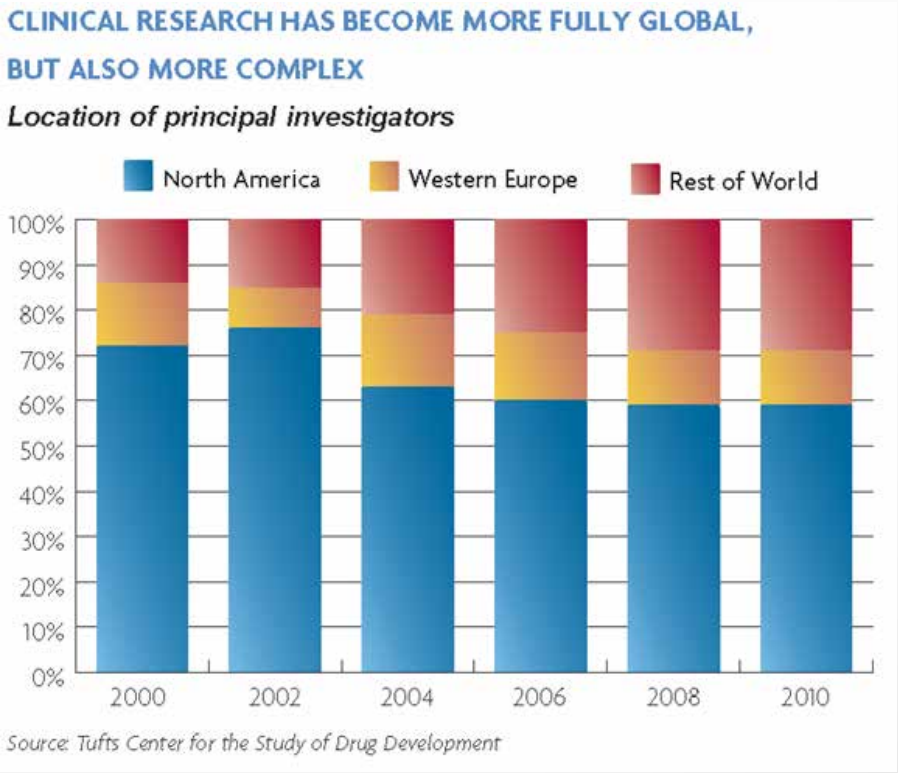
Substantiating eTMF Savings

The Pain: Paper-Based Study Costs

Two of the primary challenges facing any clinical operations manager are the sheer logistics and the costs associated with managing the mountain of essential and regulatory documents required to conduct a successful study. In order to comply with FDA, EMA, or other country-specific regulatory requirements, a pharmaceutical company or contract research organization (CRO) is required to acquire, sign, classify, and retain multitudes of compliance documents. These include 1572s, principal investigator CVs, IRB approval documents, protocol amendments, and Serious Adverse Event reporting forms, to name only a few.

Over the course of a multi-year global study, pharmaceutical, biotechnology, and medical device companies expend innumerable hours and waste a great deal of energy tracking paper documents. Additionally, companies are required by law to maintain these documents for years, further inflating the cost of compliance and drug delivery. Even setting aside the costs, the inefficiency of paper handling processes potentially slows drug development, which results in a longer time-to-market for lifesaving therapies.

A more global — and therefore more complex — clinical research landscape heightens the management burden on developers. To better manage the costs associated with global development, organizations are increasingly seeking strategies to increase efficiency. Cloud based eTMF systems can automate principal investigator enrollment, reducing study startup time and costs.



For many of us, this accumulation of paper is anathema to the sleekness of the always-on, mobile, and digital nation. We've grown accustomed to banking, shopping, paying taxes, consulting healthcare portals, and collaborating—all online. Even the federal government has mandated a "Cloud First" approach to procuring IT technology, stating on its CIO.Gov website, "The Federal Government's responsibility is to achieve the significant cost, agility and innovation benefits of cloud computing as quickly as possible." Regrettably, clinical operations has been largely stuck, operating in a paper-based model. The cost of cumbersome paper processes in the clinical space translates into higher-cost drugs. As a result, patients suffer, and drug companies lose revenue.

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100% Paperless – The SureClinical™ Vision

At SureClinical™, we pioneered the first FDA 21 Part 11 Certified Cloud™ to automate and streamline management of regulatory documents for the life sciences and healthcare industries. Our solution is purpose-built to eliminate paper at the point of origination. Coupled with our high-trust mobile digital notary service, SureEsign™, SureTrial enables the clinical operations manager to handle the entire lifecycle of a clinical document, from acquisition to completion, signing, classifying, auditing, and archiving of regulatory documents, in a 100 percent paperless environment.

In our tireless effort to eliminate paper, SureClinical has come to understand the direct and labor-related costs of administering paper processes for clinical trials. The purpose of this white paper is to equip the clinical trial manager with a real-world understanding of these costs, so that the benefits and immediate cost savings associated with adopting an eTMF can be presented to executive management and IT stakeholders. This paper examines the study characteristics, direct costs, and labor-based costs of conducting a paper-based clinical trial.

Study Characteristics

Before a clinical operations manager can begin to quantify the total cost of a study, she needs to perform a full inventory of the cost components, including the following elements (this is not an exhaustive list):

- Number of domestic and international sites
- Estimated number of documents and average page count per document
- Annual cost per square foot of storage space
- Cost of both domestic and international overnight mail
- Paper handling strategies – including scanning, faxing
- Audit frequency and anticipated staffing
- Average labor rates for scanning, tagging, filing, and hourly utilization per site
- Travel and expense cost for hotels, flights, rental cars, meals
- Document retention period (in years)

The study characteristics impact the direct costs and difficult-to-quantify labor costs in particular ways.

A single clinical trial can cost $100 million at the high end, and the combined cost of manufacturing and clinical testing for some drugs has added up to $1 billion.

—The Truly Staggering Cost of Drug Development, Matthew Harper, Forbes Magazine, 2013

Direct Costs

Direct costs are the easiest to quantify for executive management. Every dollar saved flows directly through to the bottom line as improved revenue. Conversely, adding costs such as labor, scanning or overnight mail directly increases costs of managing a paper trial master file. Below we quantify the direct and indirect costs of maintaining and managing paper-based trial master files.

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Investigator Sites

While not an immediate cost, the number of domestic and international sites underpins almost all of the costing calculations. In order to determine the total required paper, Clinical Operations needs to quantify the number of documents that are going to be disseminated to the different sites. In order to determine airway bills, the expected postage cost is determined by the number of sites. Additionally, the number of sites plays an important role in shaping the anticipated spend for hours necessary to resolve document discrepancies, to tag documents, and to file them.

Scanners

Industrial scanners are expensive. Purchasing a system that processes 50 pages per minute (PPM), with networking accessories and increased memory, can cost more than $5,000. An industry-grade scanner that processes 150 PPM with high-capacity feeder trays of more than 500 documents can cost more than $20,000 to purchase outright.

Typical leased scanners can cost as much as $1,000 a month. High-capacity toner cartridges that yield up to 10,000 pages can cost more than $200 per cartridge. Applying simple math, leasing just one industrial scanner with accessories could cost nearly $14,000 a year. Over the life of a study, the cost of maintaining that same scanner through service calls, maintenance contracts, replacement kits, and toner cartridges can easily add up to tens of thousands of dollars. Eliminating the need for scanning processes can significantly reduce the cost of a study.

Filing Cabinets

In a representative office environment, filing cabinets are not normally considered a significant capital investment. However, for clinical trials, the FDA mandates fireproof file cabinets, adding dramatically to both cost and weight. Just one three-drawer standard fireproof cabinet can cost up to $2,000 and weigh as much as 700 pounds. In an oncology study, the biopharma might generate several hundred thousand pages of documentation and medical images.

Paper Storage Costs

The cost of paper storage is not trivial. We've heard from one pharmaceutical that had to reinforce its subflooring to support the load associated with its 700-pound fireproof cabinets and the weight of paper contained therein.

However, it's probably even simpler to talk about the cost of the physical space necessary to maintain regulatory documents. Of course, the cost of space varies widely, from premiums paid in high-rent areas such as San Francisco, New York, and London, to lower-cost suburban office space. Rents can range from as high as $500 per square foot per year in the high-rent districts, to $25 per square foot in low-rent locations.

> "For investigational drug studies, clinical investigators must retain study records for a period of two years following the date a marketing application is approved for the drug for the indication for which it is being investigated..."
>
> *—Information Sheet Guidance for IRBs, Clinical Investigators, and Sponsors, FDA Inspections of Clinical Investigators, Office of Good Clinical Practice, US FDA*

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Doing the math, we see that a typical filing cabinet uses about 8 square feet (18" X 64") to accommodate being able to open the cabinets, and to meet fire code. Assuming a middle-of-the-road rent of $75 per square foot, one file cabinet costs about $600 per year to maintain. From the example in the previous section, rent is $5,400 per year. Extrapolating to ten years (eight years for drug development and an additional two years from commercial sale of the drug), the biopharma company will, on average, spend $54,000 for rental space.

Overnight Mail

Since time and security are both critical in the biopharma industry, virtually all physical documents are sent via overnight mail. Consequently, the single largest direct-cost element in a clinical trial is overnight mail. Typical overnight rates for international two-day delivery are upwards of $85 per document. Standard domestic overnight rares are about $20. Continuing with the example of a few hundred sites, and assuming that half of the study's documents are sent overnight, Clinical Operations is faced with a hefty bill of more than $215,000 per year for international overnight mail costs, and more than $50,000 for domestic overnight delivery. By the time a study is concluded, overnight mail can easily account for more than $750,000 of the total budget.

Travel, Meals, and Lodging

In a paper-based environment, the study sponsor periodically audits a site or the CRO to ensure that documents are complete and accurate. Travel budgets can run into the tens of thousands over a multi-year study, assuming quarterly travel. In order to generate a more detailed understanding of that cost, Clinical Operations will need to estimate the price tag for flights, hotels, meals, rental cars, and average time spent per quarter auditing various locations. Assuming three days per of travel quarter, modest travel expenditures, and three auditors, Clinical Operations should minimally budget $35,000 per year for the travel. This includes the cost of audit labor. Of course, these figures will vary widely based on travel locations and rates.

Interested in a free estimate of the paper tmf costs for your study?

Contact SureTrial to have one of our consultants perform a custom estimate of your study's paper-based costs compared to your estimated eTMF costs.
Call (916) 265-2008 or email info@sureclinical.com

Paper

Paper is often the least costly element of a study's overall expenses. However, during the course of a trial, with paper at $5 per ream, Clinical Operations will spend $10,000 for a million pages of printed paper.

Labor Costs

While the budget for personnel is the most expensive element of a study, the extent of labor savings resulting from the introduction of an eTMF is still somewhat difficult to calculate. At SureClinical, we use algorithms to determine the level of process automation and corresponding cost elimination. Next, we calculate the different categories of labor that are eliminated as result of adopting the various SureTrial plans. We then add back ongoing system labor to get a realistic view of overall savings.

In all cases, labor is the single biggest contributor to the total cost of the clinical study. In the examples below, only the costs for the various labor categories are calculated. The reader can then determine the percent of labor that will be saved as a result of adopting an eTMF. Labor categories that are relevant to paper-handling costs include document handling, scanning, document tagging and filing, and reporting.

Document Handling

Document handling includes management, preparation, printing, and shipping of documents. Typically, clinical operations can expect one full-time equivalent (FTE) to spend about four to five hours per month in document handling. Many of these individuals are TMF document consultants. For the purpose of these calculations, SureClinical assumed a $95 per hour rate for this category. For a study comprising a few hundred sites, the clinical operations manager can anticipate spending at least $1M per year to manage documents.

Scanning

Not only is the cost of capital investment in scanners sizable, but the labor to maintain and process scanned documents is even greater! Scanning includes the labor to scan signed documents into a digital archive or to scan documents for email distribution and digital signing. Assuming a $50 per hour labor rate, 4 hours per month per site, and the same study characteristics as above, the total budget will run approximately $480,000 per year for scanning labor.

Document Tagging and Filing

Document tagging and filing includes assigning the correct TMF naming conventions to the files, and then physically filing essential regulatory documents in fireproof cabinets. Given the same rate of pay as scanning, and assuming that clinical operations personnel spend just an hour per month filing at each site, the cost of document tagging and filing can be as much as $120,000 per year.

Report Preparation

The final key area of administrative labor includes the generation of reports to show documentation completion status, to resolve discrepancies, and to provide other key process indicators (KPIs) necessary for completion of a successful study. For the purpose of this calculation, we assumed a rate of $95 per hour for a seasoned auditor to generate and manage audit reports. At that rate, clinical operations can expect to spend as much as $450,000 per year in reporting costs.

Total Direct Costs - Paper TMF

So what is the total annual spend to support a paper-based clinical trial? Incredibly, a large global study can easily spend nearly $3.5M per year administering paper processes. More amazing is that converting to an eTMF that automates the totality of the regulatory content management can eliminate the vast majority of that amount. For any cost-conscious executive, returning three and a half million dollars to the research and development budget is the obvious choice.

Total Direct Costs - Assume 200 Global Sites:

- *Document Handling, Overnight Mail*
- *Scanning, Printing, Storage, Other*
- *Labor - Tagging and Classification*
- *Report Creation*
- *Audit travel expenses*

$3.5M / year

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SureTrial eTMF Certified Cloud™

The SureClinical SureTrial™ and Certified Cloud™ are HIPAA compliant and FDA Part 11 validated solutions that automate the acquisition, classification, signing, auditing, and archiving of clinical study regulatory documents. An efficient Software as a Service (SaaS) solution, SureTrial can be activated in minutes. The SureTrial solution generates an immediate ROI by eliminating paper-based processes, saving valuable time, energy, and dollars.

Free eTMF Return on Investment Consultation

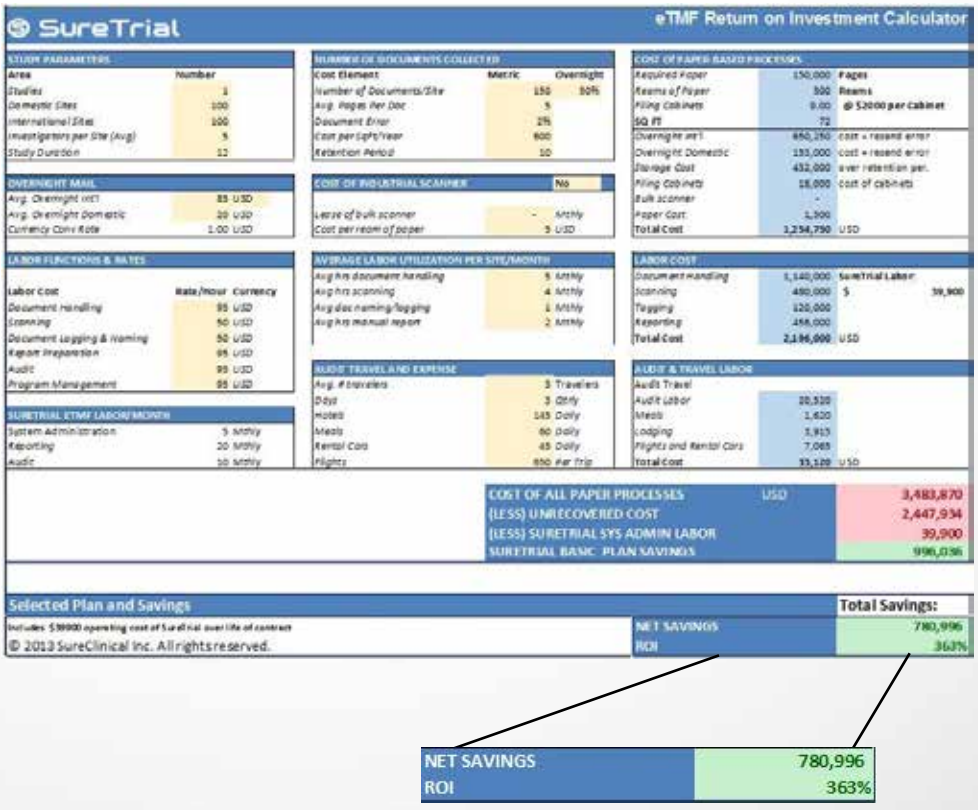
Our team of experts is prepared to work with you to estimate your current TMF costs and the estimated savings and total return on investment using the SureTrial eTMF Certified Cloud. Take advantage of this valuable consultation by contacting us today!

Contact SureClinical for an estimate of your study's paper-based costs. Learn how you can increase efficiency and lower study costs with an eTMF.

Call (916) 265-2008 or email:

info@sureclinical.com

SureTrial eTMF ROI Calculator



—Estimate your paper TMF costs

—Estimate Net Savings with SureTrial eTMF Cloud

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